

Yanming Xiu

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EDUCATION

Duke University

Aug. 2022 – Present

Ph.D. Student, Electrical and Computer Engineering

Durham, NC, USA

- **Research Interests:** Augmented Reality, Scene Understanding, Machine Learning, Computer Vision
- **GPA:** 3.86/4.0

Zhejiang University

Sep. 2018 – Jun. 2022

Bachelor of Engineering in Automation and Robotics (Chu Kochen Honor Class) Hangzhou, Zhejiang, China

- **GPA:** 3.85/4.0

RESEARCH

Generative AI-Driven Task Detrimental Content Detection in Augmented Reality

Jan. 2024 – Present

Intelligent Interactive Internet of Things Lab

Duke University

- Developed **ViDDAR**, an **obstruction attack** detection system for AR applications that prevents important real-world objects or information from being blocked by virtual content, whether intentionally or unintentionally provided by AR system. The system integrates a vision-language model (VLM), a multimodal object detection model and a segmentation model, achieving 92.15% detection accuracy with a latency of 533 ms. The resulting paper was accepted by **IEEE VR 2025** and selected as a special issue of **IEEE TVCG**.
- Systematically defined **visual information manipulation attacks** in AR through a proposed attack taxonomy, consisting of two dimensions: attack format and attack purpose. Created AR-VIM, a dataset of 452 paired raw and AR videos captured on AR devices in real-world environments. Also developed VIM-Sense, a system that combines VLMs and optical character recognition module to detect such attacks, achieving 88.94% detection accuracy. The resulting paper was accepted by **IEEE ISMAR 2025** and selected as a special issue of **IEEE TVCG**, and received the **IEEE ISMAR Best Paper Award (Top 1%)**.
- Deployed and tested both ViDDAR and VIM-Sense on android smartphones and Meta Quest 3. The work was demonstrated at **IEEE VR Research Demonstration 2025**, **The NSF Athena AI Institute for Edge Computing Summit 2025**, and multiple important internal events.
- Formally defined **contradictory virtual content attacks** in AR and proposed ContrAR, the first benchmark for evaluating VLMs under such attacks, consisting of 312 real-world AR videos collected on Meta Quest 3 and validated by human subjects. Conducted a systematic evaluation of 11 commercial and open-source VLMs, revealing clear accuracy-latency trade-offs and current limitations in semantic contradiction reasoning for AR safety. The resulting paper was accepted by the findings track of **CVPR**.
- This research direction has been presented to representatives from the U.S. Food and Drug Administration (FDA) in a research discussion led by Prof. Maria Gorlatova.

Automated Assessment Methods for Virtual Content in Augmented Reality

Nov. 2024 – Present

Intelligent Interactive Internet of Things Lab

Duke University

- Contributed to the design and construction the **DiverseAR** dataset (298 images) and its large-scale extension **DiverseAR+** (1,405 images), among the first benchmarks to systematically capture human-perceived AR visual quality factors across devices and real-world environments; additionally led the construction of **RateAR** dataset (112 videos), a video dataset for detailed virtual content visual quality assessment. All the images and videos in these datasets were captured on AR devices in real-world environments.
- Established that off-the-shelf commercial VLMs can reliably *perceive and describe* virtual content in complex AR scenes, while revealing strong dependencies on model choice and prompt design; these findings motivated a hybrid cloud-edge AR quality assessment framework. The preliminary findings of this work was accepted by **IEEE VR GenAI-XR workshop**, and the full paper was accepted by **IEEE Internet Computing, Special Issue on Immersive Computing**.

- Further demonstrated that modern VLMs can serve as human-aligned evaluators of AR content's visual quality, achieving up to 0.89 Spearman correlation with expert human judgments across placement plausibility, size appropriateness, and shadow realism, and outperforming state-of-the-art image/video quality assessment baselines. These capabilities were further translated into a closed-loop, self-improving AR content adjustment system validated by a 21-participant user study, where 19 users reported improved AR content coherence. The resulting paper is currently under review at the **IEEE ISMAR**.

Speech-Driven 3D Content Generation and Evaluation for Augmented Reality

May 2025 – Oct. 2025

Intelligent Interactive Internet of Things Lab

Duke University

- Designed a modular, edge-assisted speech-to-3D generation system for AR, supporting both direct text-to-3D and text-image-to-3D pipelines, and deployed it on a Meta Quest 3 headset with an edge GPU server.
- Implemented and systematically evaluated four representative generative pipelines via an IRB-approved user study and system-level analysis, revealing that image-guided 3D generation pipelines significantly improve perceptual quality of the generated 3D objects, while users tolerate higher latency when output quality is high. The work was accepted by the **IEEE ISMAR UNAI Workshop** and received the **Best Paper Award**.

PUBLICATIONS

Journal Publication

- **[InternetComputing'25]** L. Duan, E. Rotondo, **Y. Xiu**, S. Eom, R. Chen, C. Li, Y. Hu, and M. Gorlatova. Probing the Augmented Reality Scene Analysis Capabilities of Large Multimodal Models: Toward Reliable Real-Time Assessment Solutions. *IEEE Internet Computing, Special Issue on Immersive Computing*, 2025.
- **[TVCG'25]** **Y. Xiu** and M. Gorlatova. Detecting Visual Information Manipulation Attacks in Augmented Reality: a Multimodal Semantic Reasoning Approach. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2025. (**ISMAR Best Paper Award, Top 1%**)
- **[TVCG'25]** **Y. Xiu**, T. Scargill, and M. Gorlatova. ViDDAR: Vision Language Model-Based Task-Detrimental Content Detection for Augmented Reality. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, 2025. (**Acceptance rate: 18%**)

Conference Proceeding

- **[SubmittedVR'27]**, **Y. Xiu**, S. Mim, A. Budathoki, Y. Ding, A. Bourgeois, and M. Gorlatova, When Eyes and Ears Disagree: Detecting Multimodal Inconsistencies in Augmented Reality. Submitted to *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2026.
- **[SubmittedVRST'26]**, L. Duan, E. Rotondo, **Y. Xiu**, S. Eom, C. Li, and M. Gorlatova, Harnessing Vision-Language Models for Perceptual Quality Assessment and Autonomous Content Adjustment in Augmented Reality. Submitted to *ACM Symposium on Virtual Reality Software and Technology (VRST)*, 2026.
- **[SOUPS'26]**, X. Wang, X. Yao, **Y. Xiu**, X. Yi, M. Gorlatova, and H. Li. Is It Real? Exploiting Virtual-Physical Discrimination Vulnerability in Mixed Reality. In proceedings of *The Symposium on Usable Privacy and Security (SOUPS)*, 2026.
- **[CVPR'26F]**, **Y. Xiu**, Z. Jiang, N. Gong, and M. Gorlatova, Benchmarking Vision-Language Models under Contradictory Virtual Content Attacks in Augmented Reality. In proceedings of *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings Track*, 2026.
- **[CVPR'26F]**, R. chen, A. Andreyev, **Y. Xiu**, J. Chilukuri[†], S. Sen[†], M. Imani, B. Li, M. Gorlatova, G. Tan, and T. Lan, Cognitive Attacks Detection in Augmented Reality (CADAR): a Neuro-Symbolic Approach with Particle Filtering on Perception Graphs. In proceedings of *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings Track*, 2026.
- **[IJCNLP-AAACL'25]** S. Mim, J. Morris, M. Dhakal, **Y. Xiu**, M. Gorlatova, Y. Ding, Can a Unimodal Language Agent Provide Preferences to Tune a Multimodal Vision-Language Model? In *Proceedings of International Joint Conference on Natural Language Processing and Asia-Pacific Chapter of the Association for Computational Linguistics (AAACL)*, 2025.

Workshop Proceeding

- **[VR'26w]** J. Lin[†], **Y. Xiu** and M. Gorlatova. User Prompting Strategies and Prompt Enhancement Methods for Open-Set Object Detection in XR Environments. In Proceedings of IEEE VR Abstracts and Workshops (VRW), 2026.
- **[ISMAR'25w]** **Y. Xiu**, J. Chilukuri[†], S. Sen[†] and M. Gorlatova. Say It, See It: A Systematic Evaluation on Audio-Based 3D content Generation Methods in Augmented Reality. In proceedings of IEEE ISMAR-Adjunct, 2025. (**Best Workshop Paper Award**)
- **[VR'25w]** L. Duan*, **Y. Xiu*** and M. Gorlatova. Advancing the Understanding and Evaluation of AR-Generated Scenes: When Vision-Language Models Shine and Stumble. In Proceedings of IEEE VR Abstracts and Workshops (VRW), 2025.

Doctoral Consortium

- **[ISMAR'25dc]** **Y. Xiu**. Toward Safe, Trustworthy and Realistic Augmented Reality User Experience. In Proceedings of IEEE ISMAR-Adjunct, 2025. (**Best Doctoral Consortium Presentation Award**)

Research Demonstration

- **[VR'25d]** **Y. Xiu** and M. Gorlatova. Vision Language Model-Based Solution for Obstruction Attack in AR: A Meta Quest 3 Implementation. In Proceedings of IEEE VR Research Demonstrations, 2025.
- **[MobiHoc'25d]** **Y. Xiu** and M. Gorlatova. Demonstrating Visual Information Manipulation Attacks in Augmented Reality: A Hands-On Miniature City-Based Setup. In Proceedings of ACM MobiHoc First Workshop on Enhancing Security, Privacy, and Trust in Extended Reality (XR) Systems, 2025.

Poster Presentation

- **[ISMAR'24p]** **Y. Xiu**, T. Scargill, and M. Gorlatova. LOBSTAR: Language Model-Based Obstruction Detection for Augmented Reality. In Proceedings of IEEE ISMAR-Adjunct, 2024.

* Indicates equal contribution;

[†] Indicates students under my mentorship

TEACHING

Teaching Assistant for ECE 687D: Theory and Algorithms of Machine Learning Aug. 2025 – Dec. 2025
Department of Electrical and Computer Engineering Duke University

- Responsibilities included (1) designing homework material, exams and quizzes; (2) conducting weekly office hours and supporting students with general questions; (3) leading weekly discussion lectures and explain the problems to students; (4) Answering the questions posted by students online.

Teaching Assistant for ECE 356: Computer Networks Aug. 2024 – Dec. 2024
Department of Electrical and Computer Engineering Duke University

- Responsibilities included setting up the course website, managing the server for homework and lab projects, supporting student learning through office hours, assisting with course assessment by copy-editing and grading quizzes and exams.

High School Research Mentor Jun. 2024 – Present
Intelligent Interactive Internet of Things Lab Duke University

- Mentored 4 students from North Carolina School of Science and Mathematics (NCSSM) on two different projects.
- Student list: Junfeng Lin (NCSSM 2025, admitted by Stanford University); Tristan LoGuidice (NCSSM 2025, admitted by University of North Carolina, Chapel Hill); Shunav Sen (NCSSM 2026, admitted by Duke University); Joshua Chilukuri (NCSSM 2026)

PROFESSIONAL SERVICE

Reviewers for Journals

- Pattern Recognition (PR)
- IEEE Network
- International Journal of Human-Computer Interaction (IJHCI)
- Neurocomputing
- Engineering Applications of Artificial Intelligence (EAAI)

Reviewers for Conferences

- ACM VRST
- EuroXR International Conference (EuroXR)

External Reviewers for Journals and Conferences

- 2026: IEEE/CVF CVPR, ACM MobiSys, ACM SenSys
- 2025: ACM SIGComm
- 2024: ACM MobiCom, ACM SenSys

HONORS

Best Paper Award <i>IEEE International Symposium on Mixed and Augmented Reality (ISMAR)</i>	Oct. 2025
Best Doctoral Consortium Presentation Award <i>IEEE International Symposium on Mixed and Augmented Reality (ISMAR)</i>	Oct. 2025
Best Paper Award <i>IEEE ISMAR Universal Augmented Interaction (UNAI) Workshop</i>	Oct. 2025
Graduate Program Research Workshop Student Talk Session - Runner Up Award <i>Department of Electrical and Computer Engineering, Duke University</i>	Sep. 2025
Outstanding Graduate (Bachelor's Program) <i>Chu Kochen Honors College, Zhejiang University</i>	Jun. 2022
Outstanding Undergraduate Student Award <i>Chu Kochen Honors College, Zhejiang University</i>	May 2021
The First Class Undergraduate Scholarship <i>Chu Kochen Honors College, Zhejiang University</i>	May 2021